

Task 1: To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

```
Radhakrishna@04:~# nano loop.c
```

```
#include <stdio.h>
#include <string.h>

int main() {
    char input[100];

    printf("=== Interactive Main Loop ===\n");
    printf("Type 'exit' to quit.\n\n");

    while (1) {
        printf("Enter a command: ");

        if (fgets(input, sizeof(input), stdin) == NULL) {
            break;
        }

        input[strcspn(input, "\n")] = '\0';

        if (strcmp(input, "exit") == 0) {
            printf("Exiting program...\n");
            break;
        }

        printf("You entered: %s\n", input);
    }
}
```

```
    }

    return 0;
}
```

```
Radhakrishna@04:~# gcc loop.c -o loop
Radhakrishna@04:~# ./loop
```

```
Radhakrishna@04:~# ./loop
=== Interactive Main Loop ===
Type 'exit' to quit.

Enter a command: hello
You entered: hello
Enter a command: linux
You entered: linux
Enter a command: ossp
You entered: ossp
Enter a command: Bye
You entered: Bye
Enter a command: exist
You entered: exist
Enter a command: exit
Exiting program...
```

Task 2: Using the following system calls, copy the content from File1.txt to File2.txt.

1. **Open()**
2. **Read()**
3. **Write()**
4. **Close()**

```
Radhakrishna@04:~# nano copycontent.c
```

```

#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main()
{
    int fd1, fd2;
    char buffer[100];
    int n;

    fd1 = open("File1.txt", O_RDONLY);

    if (fd1 < 0)
    {
        perror("Error opening File1.txt");
        return 1;
    }

    fd2 = open("File2.txt", O_CREAT | O_WRONLY | O_TRUNC, 0644);

    if (fd2 < 0)
    {
        perror("Error opening File2.txt");
        close(fd1);
        return 1;
    }

```

[Read 40 lines]

```

    }

    fd2 = open("File2.txt", O_CREAT | O_WRONLY | O_TRUNC, 0644);

    if (fd2 < 0)
    {
        perror("Error opening File2.txt");
        close(fd1);
        return 1;
    }

    while ((n = read(fd1, buffer, sizeof(buffer))) > 0)
    {
        write(fd2, buffer, n);
    }

    printf("Contents copied successfully from File1.txt to File2.txt\n");

    close(fd1);
    close(fd2);

    return 0;
}

```

```
Radhakrishna@04:~# echo "Hello Linux System Calls" > File1.txt
Radhakrishna@04:~# gcc copycontent.c -o copycontent
Radhakrishna@04:~# ./copycontent
Contents copied successfully from File1.txt to File2.txt
Radhakrishna@04:~# cat File2.txt
Hello Linux System Calls
Radhakrishna@04:~# nano copycontent.c
Radhakrishna@04:~# |
```